

"Debate Coach" Bias & Persuasion Analysis

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Fostering critical thinking through AI conversation

1 Introduction

Virtual communities are often echo chambers, and consumers rarely engage with counterarguments. Debate Coach uses AI to create a "devil's advocate" (rather than a winning debater) whose job is to strengthen the user's argument. Participants can select a topic, debate with the AI about it, and afterward reflect on whether the AI challenged their thinking or exposed a blind spot. Or feel it's unfair?

We then compare a base language model to one that has been fine-tuned on high-quality transcripts of debates. This acts as a case study of a common problem in human-AI interaction, specifically how to get an AI to constructively interrogate a user's beliefs without triggering defensiveness. We conduct a small user study measuring stance reconsideration, perceived fairness, and perceived frustration to investigate how to train AIs to promote open-mindedness, empathy, and civil discourse.

2 Research Problem

The research problem we aim to address is how AI systems can effectively challenge users' beliefs without causing defensiveness or disengagement. In many online environments, individuals are exposed primarily to information that reinforces their existing viewpoints, limiting opportunities for critical thinking. While AI has the potential to introduce counterarguments, poorly designed systems may come across as hostile, biased, or unhelpful. This raises the question of how to design AI systems that can promote thoughtful reflection while maintaining a sense of fairness and engagement.

3 Research Context and Problem Statement

4 Proposed Solution

Our approach is to develop and evaluate a system called "Debate Coach," which uses AI to act as a constructive devil's advocate. Instead of trying to "win" arguments, the AI is designed to strengthen the user's reasoning by presenting thoughtful counterpoints [1, 2]. We compare a baseline language model with versions that are prompted and fine-tuned on high-quality debate data. Additionally, we incorporate user feedback and evaluation metrics to refine how the AI presents arguments. This approach aims to balance challenge and support, encouraging users to reconsider their perspectives without feeling attacked.

5 Evaluation and Implementation Plan

5.1 Evaluation Plan

We evaluate our three conditions using a mixed evaluation of computational metrics and an informal human pilot study. The next sections describe the specifics of these metrics and study.

- **Condition A** (vanilla)
- **Condition B** (prompted devil's advocate)
- **Condition C** (adapted devil's advocate)

In terms of technical and UX implementation.

5.1.1 Automated Evaluation

- **LLM-as-Judge** Building on a recent LLM-based assessment study [3], we use GPT-4 to rate the transcripts in each condition for five properties:

1. strength of counterargument
2. engagement with the user's true arguments
3. fairness (absence of straw-manning)
4. persuasive appeal
5. constructive appeal.

GPT-4 also provides justifications for its ratings. For all five properties, a rating of 1-5 is assigned. Scores are averaged across transcripts per condition to form a comparative profile.

- **Argument Quality Metrics** We also derive the following automated metrics from all debate transcripts:

1. **Argument Diversity** The cosine distance between successive argument embeddings, with a lower value indicating more argument diversity.
2. **Topical Relevance** The cosine similarity between the terms in user messages and the ones in the AI response is a measure of topical relevance.
3. **Repetition Rate** N-gram overlap across the AI’s turns within a single debate. The less overlap, the less recycling of arguments.

5.1.2 Informal Pilot Study

Continuing, we again run a small within-subjects pilot study with 5-10 participants (drawn from our academic network) to choose from a list of somewhat controversial topics that they would like to debate in a 5-7-minute format with each of the three conditions. The condition order is counterbalanced between subjects to control for order effects.

At the conclusion of each debate, participants were asked to indicate on a 1-5 Likert scale four traits:

1. **Stance Reconsideration** If the AI prompted the user to rethink their belief.
2. **Perceived Fairness** Did the AI argue in good faith and engage with the user’s actual claims?
3. **Helpfulness** The extent to which the interaction helped the user understand the opposing view.
4. **Frustration** Whether the AI was pushy, repetitive, or hostile.

5.1.3 Analysis

At the end, participants evaluate each condition, saying which was the hardest and most fair and which would be chosen again if the experiment were run a second time.

The mean values of automated metrics and the respective LLM-as-Judge scores by condition are reported and presented as bar plots for comparison. In the pilot study, descriptive measures (means and standard deviations) are reported for main survey dimensions, and a lightweight thematic analysis is provided for open-ended survey responses across the four dimensions. Because our sample is small, we do not run inferential statistics on our data and present our results as suggestive evidence for a larger controlled study.

5.2 Timeline

The 10-week period has four phases with final deliverable, as shown in Table 1, with each phase building on the previous one. In the first phase (weeks 1-2), the

project is set up. The student sets up a repository, reads the relevant literature, and has an initial discussion with the mentor to confirm/define scope and expectations. Weeks 3-6 are the "core" development sprint, where we implement all three chatbot conditions plus the debate UI. These four weeks are the longest sprint because they involve the most technical risk, especially the condition C adaptation work. Thus, weeks 7 and 8 are reserved as a buffer for iteration and study preparation, such that we can make changes to our prototype based on internal testing and finalize our evaluation instruments before working with participants. Weeks 9-10 consist of conducting an informal pilot with 5-10 users, followed by a quick analysis of the findings. The second deadline, on April 29, leaves a week for report writing, repository cleaning, and demo preparation.

It is structured such that even if later stages are delayed, a prototype and a study design are still produced for use as minimum deliverable outcomes.

Table 1: Project Timeline

Phase	Dates	Weeks	Activities
Onboarding + Setup	Feb 17 – Feb 28	1–2	Repo, reading, corpus sourcing, mentor conversation about scope
All 3 Conditions + UI	Mar 1 – Mar 28	3–6	Condition A (vanilla), B (prompted), C (adapted), chat UI, reflection flow
Iteration + Study Design	Mar 29 – Apr 11	7–8	Polish prototype, draft survey for internal evaluation, build analysis notebook templates
Informal Pilot	Apr 12 – Apr 22	9–10	Run 5–10 people through all 3 conditions, collect data, quick analysis
Final Deliverable	Apr 29	Deadline	Report, demo, repo, pilot dataset

6 Conclusion

7 Revision Edits

Added sections, lists, tables, and citations to demonstrate our understanding of LaTeX and Overleaf.

8 References

- [1] Tianyu Hu, Zhen Tan, Song Wang, Huaizhi Qu, and Tianlong Chen. Multi-agent debate for llm judges with adaptive stability detection. *arXiv preprint arXiv:2510.12697*, <https://arxiv.org/abs/2510.12697>, 2025.
- [2] Owen Rambow, Leo Wanner, Marianna Apidianaki, Hend Al-Khalifa, Barbara Di Eugenio, and Steven Schockaert. Debate-to-write: A persona-driven multi-agent framework for diverse argument generation. *arXiv preprint https://aclanthology.org/2025.coling-main.314/*, 2025.

- [3] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. *arXiv preprint arXiv:2306.05685*, <https://arxiv.org/abs/2306.05685>, 2023.